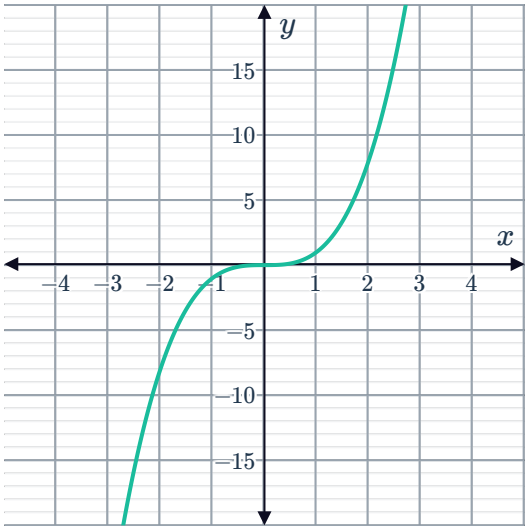
Shape of cubics

1 Is the function $y = -2x^3 - 4$ one-to-one?

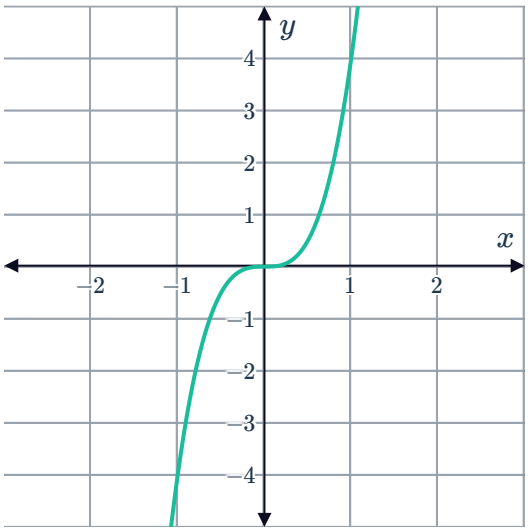
2 By considering the graph of $y = x^3$, determine the following:

a As x approaches infinity, what happens to the corresponding y-values?

b As x approaches negative infinity, what happens to the corresponding y-values?



3 Determine whether the graphed function shown has an even or odd power. Explain your answer.



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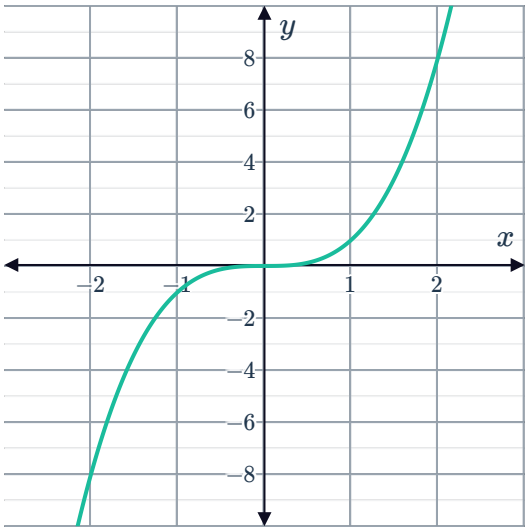
4 Consider the graph of the function $y = x^3$:

a For what values of x is the functions concave down?

b For what values of x is the functions concave up?

c At what point does the concavity of the curve change?

d What is this point called?

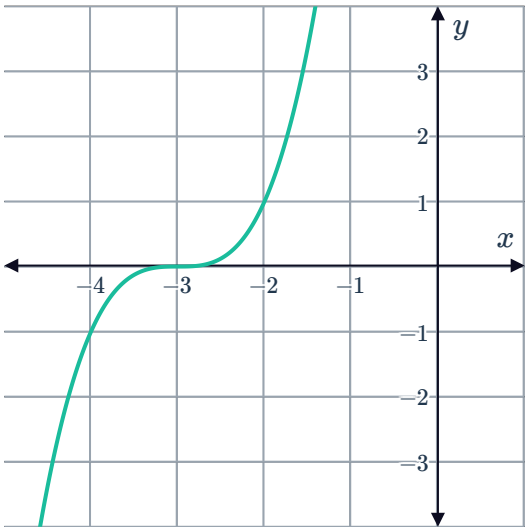


5 Consider the graph of the function:

a For what values of x is the cubic concave up?

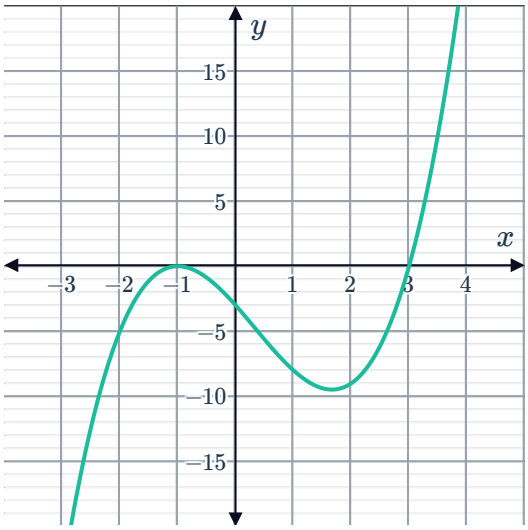
b For what values of x is the cubic concave down?

c State the coordinates of the point of inflection.



6 How does the graph of $y = \frac{1}{2}x^3$ differ to the graph of $y = x^3$?

7 Consider the graph of the cubic function shown. For which values of x is $y \geq 0$?



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8 The graph of $y = x^3$ has a point of inflection at $(0, 0)$. By considering the transformations that have taken place, find the point of inflection of each cubic curve below:

a $y = \frac{2}{3}x^3$ **b** $y = x^3 + 3$ **c** $y = -x^3 + 4$

Finding intercepts and equations of cubics

9 Consider the graph of the function with equation of the form $y = ax^3 + bx^2 + cx + d$:

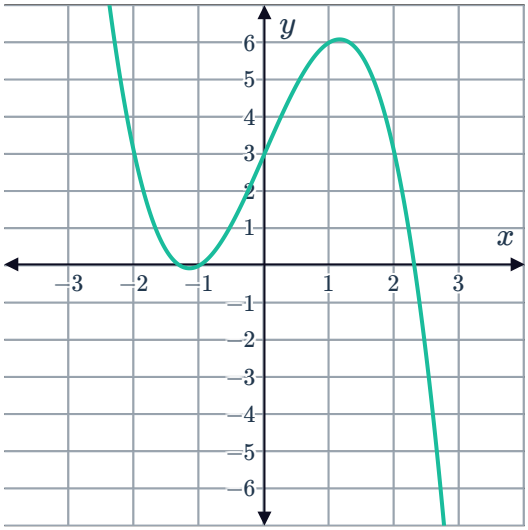
a Is the value of a is positive or negative?

b State the coordinates of the y-intercept.

c For which values of x is the graph concave up?

d For which values of x is the graph concave down?

e State the coordinates of the point of inflection.



10 State whether the following functions pass through the origin.

a $y = (x-2)^2(x+3)$ **b** $y = (x+1)^3$

c $y = (x-4)(x+7)(x-5)$ **d** $y = x(x-6)(x+8)$

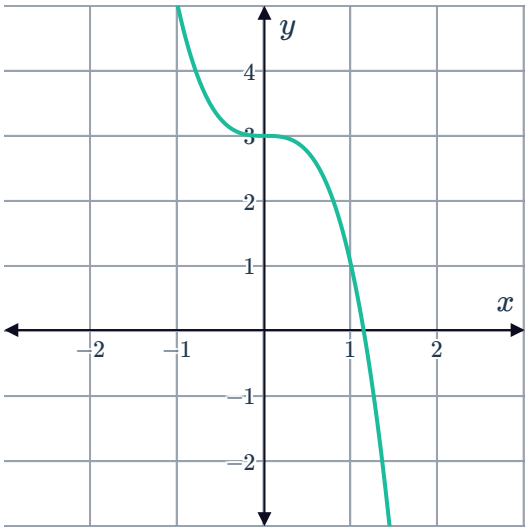
11 A cubic function is defined as $y = \frac{1}{2}x^3 + 4$.

a Find the x-intercept of the function. **b** Find the y-intercept of the function.

12 A cubic function has the equation $y = x(x-4)(x-3)$. How many x-intercepts does it have?

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13 Consider the given graph of a cubic function:



a Determine whether the cubic is positive or negative.

b State the coordinates of the y-intercept.

c Which of the following could be the equation of the function?

A $y = -2x^3 + 3$ **B** $y = 2x^3 + 3$

C $y = 2x^3 - 3$ **D** $y = -2x^3 - 3$

14 If the graph of $y = x^3$ is moved to the right by 10 units, what is the new equation?

Graphs of cubics

15 Consider the cubic function $y = -x^3$

a Complete the following table of values.

x	-2	-1	0	1	2
y					

b Sketch the graph of the curve.

16 Consider the curve $y = x^3 - 8$.

a Find the x-intercept.

b Find the y-intercept.

c Find the horizontal point of inflection.

d Sketch the graph of the curve.

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17 For each of the functions below:

i Is the cubic increasing or decreasing from left to right?

ii Is the cubic more or less steep than the cubic $y = x^3$?

iii What are the coordinates of the point of inflection?

iv Sketch the graph.

a $y = 2x^3 + 2$

b $y = -\dfrac{x^3}{4} + 2$

c $y = -2\,(x-2)^3$

d $y = \dfrac{1}{2}\,(x-3)^3$

e $y = -\dfrac{1}{2}\,(x-1)^3 + 2$

18 Consider the curve $y = -3\,(x-1)^3 + 3$.

a Find the x-intercept.

b Find the y-intercept.

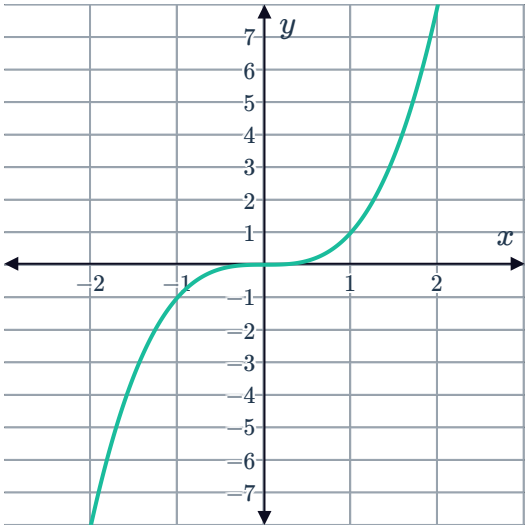
c State the point of inflection.

d Sketch the graph of the curve.

19 Consider the graph of $y = x^3$.

a How do we shift the graph of $y = x^3$ to get the graph of $y = (x-2)^3 - 3$?

b Hence plot $y = (x-2)^3 - 3$.



20 Sketch the function $y = (x-2)\,(x+1)\,(x+4)$ showing the general shape of the curve and the x-intercepts.

21 For each of the following curves:

i Find the x-intercept(s).

ii Find the y-intercept(s).

iii Sketch the graph of the curve.

a $y = (x+3)\,(x+2)\,(x-2)$

b $y = -(x+4)\,(x+2)\,(x-1)$

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22 Consider the equation $y = 3x + 2x^2 - x^3$.

a Express the equation in factorised form.

b Find the y-intercept.

c Find the x-intercepts.

d Sketch the graph of the curve.

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